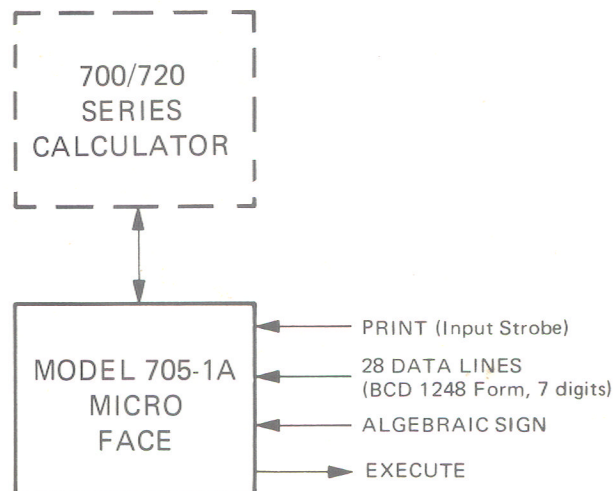
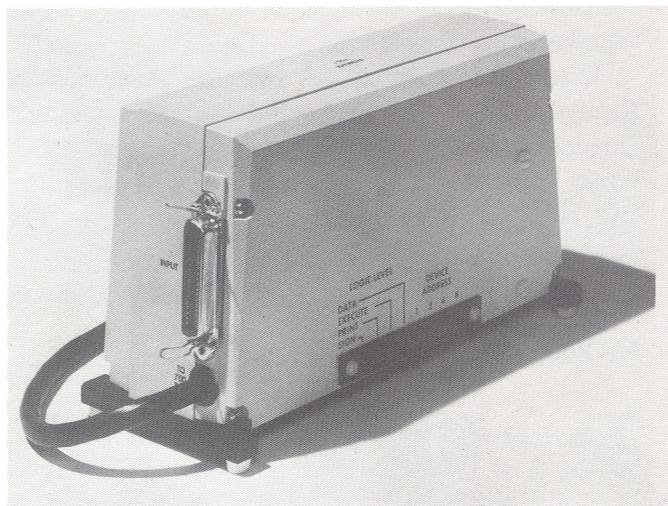


WANG MODEL 705-1A MICRO FACE



GENERAL INFORMATION

The Model 705-1A Micro Face allows a variety of external digital devices (digital voltmeters, panel meters, counters, digital clocks, etc.) to be interfaced to the Model 700 Series Programmable Calculators for many on-line applications. As an input-only interface, the Model 705-1A can easily accept numeric data of up to seven digits plus algebraic sign in parallel using standard BCD 1, 2, 4, 8 format. The Micro Face consists of an input connector, a code converter which converts standard BCD 1, 2, 4, 8 codes into the hexadecimal codes required for the programmable calculator language, a parallel-to-serial converter, and two control signals. Design conveniences of this interface enables adaption to fixed or floating point entry for automatic ranging.

INPUT DATA SIGNAL LEVELS

Input circuits are TTL/DTL compatible. (See Figure 1 for typical input circuit.) Voltage levels for a logic "0" are between 0 to +0.4 volts and voltage levels for a logic "1" are between +2.5 to 5 volts. A convenient slide switch is located on the

rear cover to reverse these logic definitions to the complement bar functions (i.e., 1 = 0 to 0.4 volts and 0 = +2.5 to 5 volts). Input impedance is typically 4K ohms with a 47K ohm resistor in parallel which requires approximately 1.6 milliamp sink (see Figure 1). The pin diagram (Figure 3) shows signal input digit positions, as well as control signals and their associated BCD codes.

CONTROL SIGNALS

The Micro Face allows for one input control signal ("Print" command) and provides one output control signal ("Execute" command).

The "Print" command, which is an input control signal, indicates that information is present on the input lines and is available for transfer. This signal should be connected to pin 18 of the input connector (Figure 3). The "Print" command (input strobe) is normally a d.c. condition of 0 or +5 volts. A convenient slide switch is available to reverse these logic levels. If a pulse is used for the "Print" command, a pulse width of approximately 5 microseconds is sufficient to be recognized by the Micro Face.

MODEL 705-1A MICRO FACE (cont.)

The "Execute" command, which is an output signal from the Micro Face, indicates that the Micro Face is ready to receive numeric data. The "Execute" command is a d.c. condition of 0 or +5 volts with a convenient slide switch to reverse logic levels. The "Execute" d.c. condition is available at pin 31 of the input connector (Figure 3). Output impedance is typically 470 ohms and provides for a 10-milliamp sink. (See Figure 2.)

SIGN OF A NUMBER

A one-bit input at pin 17 of the input connector (Figure 3) is used to designate the sign of the number. Signal levels are the same as those of the data input signals. A convenient slide switch is available to reverse the logic levels of the sign bit.

FIXED DECIMAL POINT

Any one of the seven (7) input digits can be wired to generate a decimal point BCD code 12 allowing for a six (6) digit number plus fixed decimal point.

EXAMPLE: If input numbers are 123.456, the fourth digit can be wired for a decimal point (BCD code 12).

FLOATING DECIMAL POINT

The last digit can be wired to indicate the number of places the decimal point should be located within a six (6) digit number.

EXAMPLE If the input number is 123456 x 10⁻³ the range digit can be wired as the 7th digit and the number would read in as 1234563 (where the 3 indicates under program control the scale factor). This feature can be used for automatic ranging.

ADDRESSING THE 705-1 INPUT MICRO FACE

To switch control from a programmable calculator, a two-step command must be issued either manually from the keyboard or as a program step in core memory.

Addressing the Micro Face requires a Group 1, 07XX (where "XX" can be any number from 00

through 15), thus allowing the operator to choose an address by setting four (4) switches. This feature also allows the use of more than one interface for multiplexing purposes by calling out different addresses. (Because of the IC fan-out restrictions, no more than 10 interfaces are recommended.)

There are eight slide switches conveniently located on the rear cover. Four of these switches (marked "Device Address") generate the "XX" codes 00 through 15 which will be recognized by the selected Micro Face when addressed. The remaining four switches provide the following flexibility as marked.

MARKINGS

Data	Reverse logic levels for input signal
Print	Reverse logic levels for "Print" command
Execute	Reverse logic levels for "Execute" command
Sign	Reverse logic levels for Sign of number

PARALLEL INPUT

The input connector is conveniently wired for direct parallel input of seven digits (28 lines as shown in Figure 3).

Typical entry sequence would be:

Group 1 07XX	2-step command from the programmable calculator transfers control to the selected input interface.
"EXECUTE" ON	D.C. condition changes at pin 31 and remains in that state waiting for a "Print" command (input strobe) from the outside device.
"PRINT"	Command generated from outside device to pin 18 for approximately 5 microseconds or greater. All seven digits are then transferred into the display register of the programmable calculator in approximately 5 milliseconds.
"EXECUTE" OFF	D.C. condition returns to original state and indicates transfer is complete and control is returned to the programmable calculator.

MODEL 705-1A MICRO FACE (cont.)

PARALLEL INPUT (Cont.)

One 7-digit number has now been transferred from the outside device into the display register of the programmable calculator.

This procedure is repeated for each new number. Transfer time of each 7-digit number is approximately 5 milliseconds.

CAUTION: *A number that has been read into the display register must be moved to another location within the calculator prior to reading in another number, otherwise the first number will be lost.*

INTERCONNECTING CABLE

A six-foot cable is provided with each micro face. Attached to one end of this cable is a 36-pin Amphenol male connector (Amphenol P/N 57-30360, Wang Labs, Inc., P/N 350-2025), which plugs into the micro face female connector marked "Input." The other end of the cable is open-ended to allow the user to attach any type of mating connector required for his own outside device. (This connector is usually available from the equipment manufacturer of the outside device.) If any questions arise, please do not hesitate to contact Wang Laboratories, Inc. in Tewksbury, Mass. for wiring diagram assistance.

NOTE: *These specifications are subject to change without notice.*

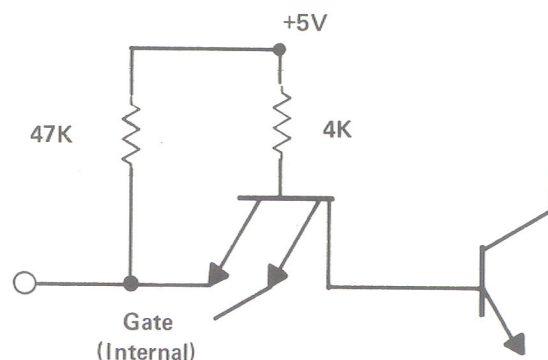


FIGURE 1 Typical Input Circuit

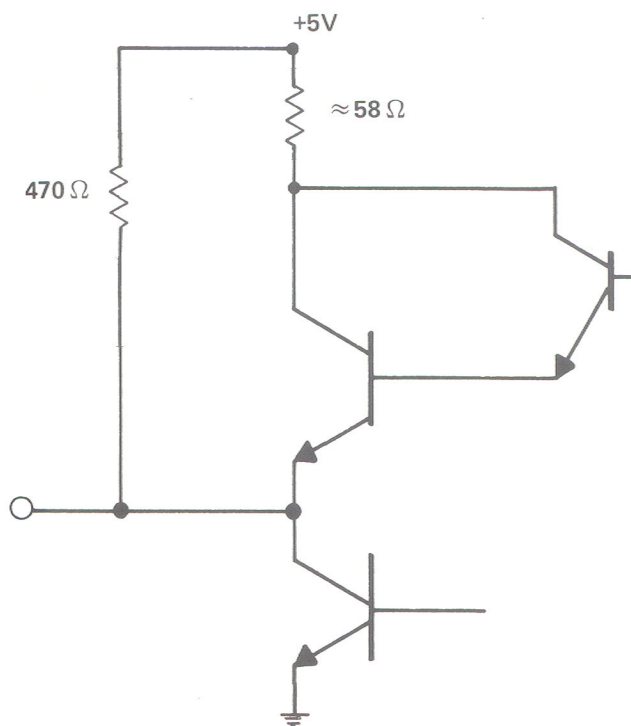


FIGURE 2 Typical Output "Execute" Circuit

MODEL 705-1A MICRO FACE (cont.)

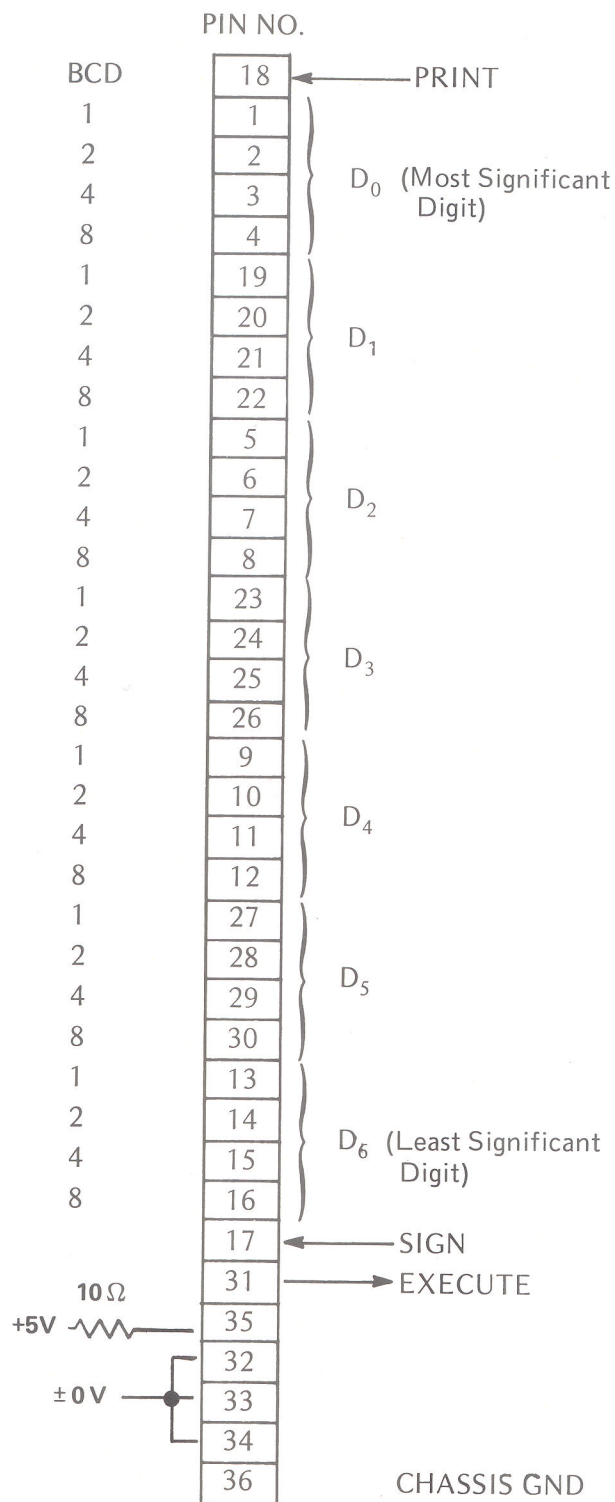


FIGURE 3 36-Pin Input Connector Pin Diagram